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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

Inventory Purchase Order Management System content

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Academic Year: **2025 – 2026**

APRIL 2026

Semester: **II**

DECLARATION

I hereby declare that the project entitled **"Inventory Purchase Order Management System content"** submitted in partial fulfillment of the requirements for the course **Java programming** is a **record of my original work** carried out under the guidance of **Mr. Dharaneesh**Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

SUPRIYA. R

CERTIFICATE

This is to certify that the project entitled **" Inventory Purchase Order Management System content"** is a **bona fide work** carried out by **SUPRIYA R, [1-'C'/2025-2026], B.Sc.Computer Science**, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **Java programming**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: Coimbatore

Date: April 2026

Submitted for the Viva-Voce Examination held on. 10/04/2026

Guide:
(Technical Trainer, KGCAS)

Mr.DHARANEESH.

Head of the Department
(Technical Head, KGCAS)

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ABSTRACT

The Inventory Purchase Order Management System is a Java-based application designed to manage inventory and streamline the purchase or

process in small and medium-scale businesses. Manual inventory handling

often leads to stock mismanagement, delayed ordering, and errors in record keeping. This system automates inventory tracking and purchase order generation, ensuring efficiency and accuracy.

The application allows users to add, update, and monitor inventory items, track stock levels, and automatically generate purchase orders when stock reaches a minimum threshold. It is developed using Core Java concepts such as classes, objects, file handling, loops, and exception handling.

The system improves inventory control, reduces human errors, and ensures timely procurement of goods. It serves as a basic foundation for more advanced enterprise inventory systems.

1. INTRODUCTION

Inventory management is a critical function in any business that deals with physical goods. Traditional methods of inventory tracking using manual registers or spreadsheets are time-consuming and prone to errors.

The Inventory Purchase Order Management System is developed to automate inventory operations such as stock tracking, updating product details, and generating purchase orders. The system helps businesses maintain optimal stock levels and avoid overstocking or stockouts.

This project demonstrates how Java programming can be used to build real-world business applications with efficient data handling and processing.

1.1 Objective

The main objectives of this system are:

To develop a Java-based inventory management system

- To track stock levels of products
- To automate purchase order generation
- To reduce manual errors in inventory handling To provide a user-friendly interface
- To demonstrate Core Java concepts

1.2 Overview

The system is a menu-driven application where users

can:

- Add new products to inventory Update stock quantities
- View available stock
- Generate purchase orders when stock is low
- The application calculates required stock automatically and generates purchase details for suppliers

1.3 Features

- Product Management Stock Tracking
- Low Stock Alert
- Purchase Order Generation Automatic Calculations Simple User Interface
- Fast Processing

2. PROGRAM ANALYSIS

2.1 Problem Definition

Inaccurate stock tracking

Delay in purchase orders Difficulty in maintaining records Overstocking or understocking Human errors

This system solves these problems by automating inventory management.

2.2 Project Overview

The system includes:

Product database management Stock monitoring

Purchase order generation

Users can manage products and ensure timely restocking through automated alerts.

2.3 Scopes and Limitations

Scope

Small businesses Warehouses Retail stores

Inventory tracking systems

Limitations

No database integration (basic version)

Limited to local system usage

No online access

3. SYSTEM DESIGN

Modules

..**Product Management.**

- Module Add new product
- Update product details Delete product

Inventory Tracking Module

- Monitor stock levels Display available products
- Update stock after purchase/sale

Purchase Order Module

- Generate purchase order
- Calculate required quantity
- Display supplier details

Alert Module

- Detect low stock Notify user for reorder

Exit Module

- Safely terminate the system

4. METHODOLOGY

4.1. Problem Analysis

Identified issues in manual inventory handling and defined system requirements.

4.2. System Design

- System divided into modules:
- Product management Inventory tracking
- Purchase order

4.3. Program Development

- **Implemented using Java:**
- **Classes and objects**
- **Loops and conditions**
- **File handling**

4.4. Testing and

Test Case	**Expected Result**
Update stock	Stock updated
Add product	Product stored correctly
Low stock	Alert generated
Generate PO	Correct order

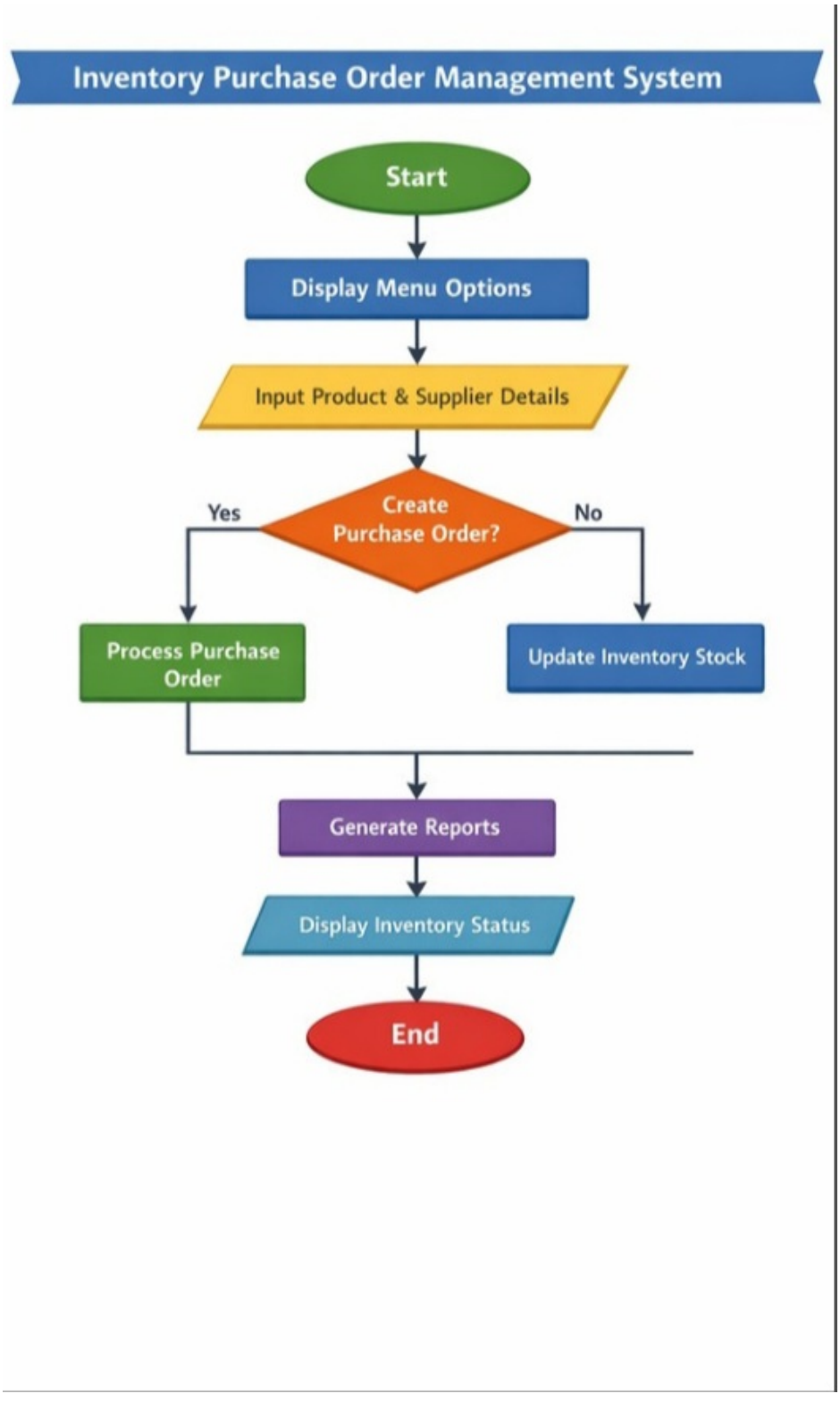
4.5. Result and Documentation

System successfully manages inventory and generates purchase orders.

4.6. Algorithm

1. Start program
2. Display menu
3. Select option
4. Add/update/view products
5. Check stock level
6. If stock is low → generate purchase order
7. Display results
8. Exit

4.7. Flowchart



5. SOURCE CODE

```
import java.util.*;
import java.io.*;

public class Main {

    // Class inside same file
    static class Product {
        int id, qty, min;
        String name;

        Product(int id, String name, int qty, int min) {
            this.id = id;
            this.name = name;
            this.qty = qty;
            this.min = min;
        }
    }

    public static void main(String[] args) {
        ArrayList<Product> list = new ArrayList<>();
        Scanner sc = new Scanner(System.in);
        int ch;
```

```
do {
    System.out.println("\n1.Add 2.View 3.Low 4.Save 5.Load 6.Exit");
    System.out.print("Choice: ");
    ch = sc.nextInt();

    switch (ch) {
        case 1:
            System.out.print("ID: ");
            int id = sc.nextInt(); sc.nextLine();

            System.out.print("Name: ");
            String name = sc.nextLine();

            System.out.print("Qty: ");
            int qty = sc.nextInt();

            System.out.print("Min: ");
            int min = sc.nextInt();

            list.add(new Product(id, name, qty, min));
            break;

        case 2:
            for (Product p : list) {
                System.out.println(p.id + " " + p.name + " " + p.qty);
            }
            break;
```

```
case 3:
    for (Product p : list) {
        if (p.qty < p.min) {
            System.out.println("Low Stock: " + p.name);
        }
    }
    break;

case 4:
    try {
        PrintWriter pw = new PrintWriter("data.txt");
        for (Product p : list) {
            pw.println(p.id + "," + p.name + "," + p.qty + "," + p.min);
        }
        pw.close();
        System.out.println("Saved!");
    } catch (Exception e) {
        System.out.println("Error saving");
    }
    break;

case 5:
    try {
        File f = new File("data.txt");
        if (!f.exists()) break;

        Scanner fs = new Scanner(f);
        list.clear();
```

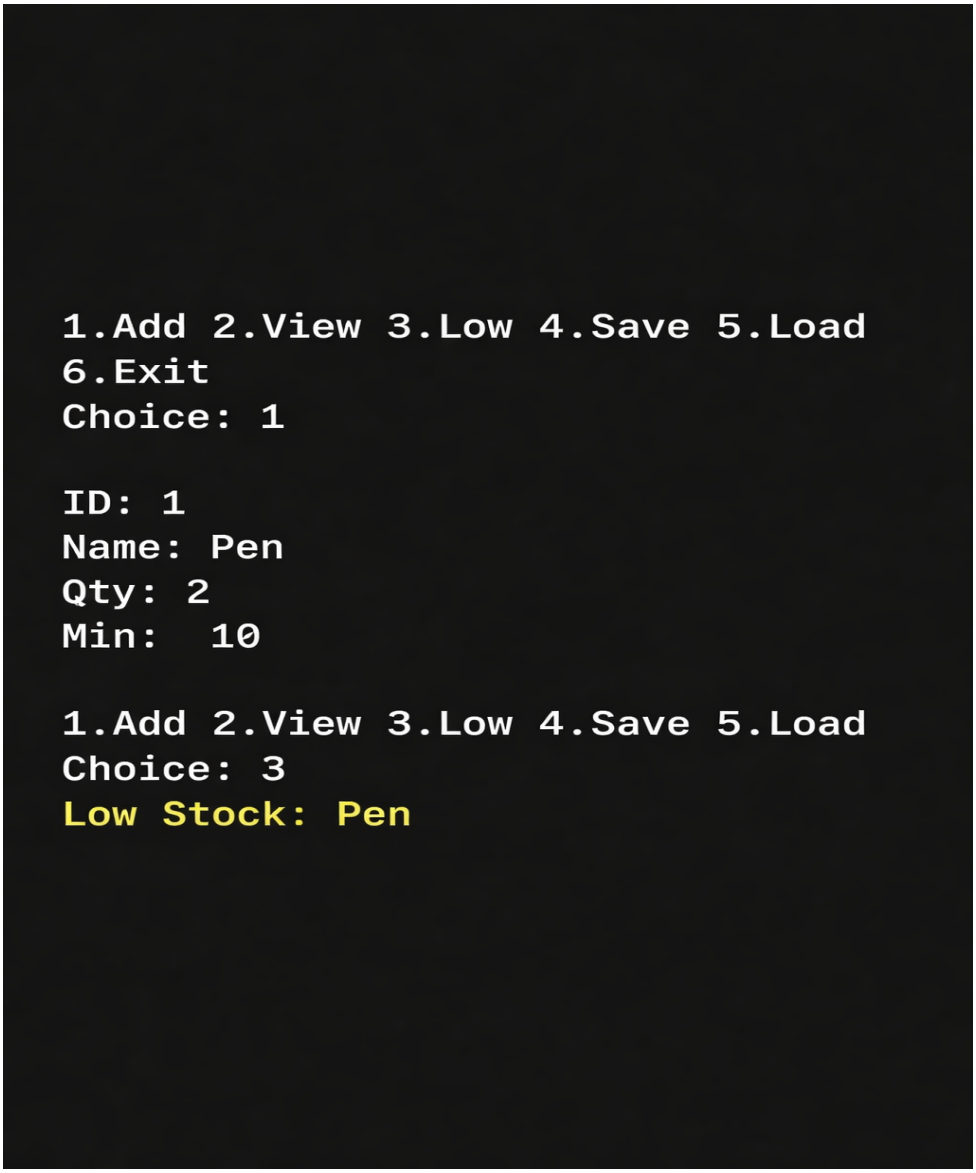
```
                while (fs.hasNextLine()) {
                    String[] d = fs.nextLine().split(",");
                    list.add(new Product(
                        Integer.parseInt(d[0]),
                        d[1],
                        Integer.parseInt(d[2]),
                        Integer.parseInt(d[3])
                    ));
                }
                fs.close();
                System.out.println("Loaded!");
            } catch (Exception e) {
                System.out.println("Error loading");
            }
            break;

        case 6:
            System.out.println("Exit");
            break;

        default:
            System.out.println("Invalid");
      }

    } while (ch != 6);
  }
}
```

6. INPUT AND OUTPUT SCREENS



7. DISCUSSION & CONCLUSION

7.1 Discussion

The system simplifies inventory management by automating stock tracking and purchase order generation. It reduces manual effort and improves efficiency.

7.2 Conclusion

The Inventory Purchase Order Management System successfully demonstrates how Java can be used to build business applications. It ensures better stock control, reduces errors, and improves productivity.

Future enhancements may include:

- Database integration
- GUI interface
- Online inventory tracking

8. REFERENCES

Books

- Herbert Schildt – Java: The Complete

Websites

- https://www.oracle.com/java
- https://www.geeksforgeeks.org
- https://www.javatpoint.com